Factoring the Counting Problem Out of Coupled Particle-Field Renormalisation

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Abstract

The Doi–Peliti generating function of a reaction network, written in the form standardly produced by the algebraic field-theoretic formalism, is already a Lagrange equation in the sense of Flajolet–Sedgewick analytic combinatorics. This recognition—stated as a theorem in the companion note (**?**)—imports four consequences wholesale: (C1) all-orders coefficient asymptotics via the transfer theorem, (C2) classification of mean-field universality by the Puiseux exponent p/q, (C3) non-perturbative nucleation rates as a branch-gap contour integral, and (C4) UV divergence structure by power counting on a finite 1PI enumeration. We call the resulting framework Coupled Field Analytic Combinatorics (CFAC). This paper elaborates (C1)–(C4) at coupled-field scope on a Doi–Peliti/Martin–Siggia–Rose Keller–Segel system and establishes three concrete results: **Result 1** one-loop perturbative exactness of the minimal coupling sector, established by exhaustive 1PI graph enumeration at $L \leq 3$—a coupled-scope sharpening of (C4); **Result 2** the branch-gap nucleation formula (C3) verified to six significant figures on seven parameter sets; **Result 3** three-loop two-colour Kirchhoff factorisation for every physically relevant edge colouring. Two supporting results extend the algebraic reach: the one-loop bridge integral has a *closed tropical form*—Arkani-Hamed–Hillman–Salvatori leading-facet-plus-Hessian—which is exact, not asymptotic, at one loop, and a spectral-dimension substitution $d \to d_s$ makes (C1) and (C2) available on fractal substrates (verified on the Sierpinski gasket to $< 2\%$). Illustrative applications to amyloid fibril nucleation and ice crystal formation yield order-of-magnitude estimates from published rate constants in regimes where direct simulation cost scales as $\exp(VS)$. Section 12.6 delimits scope: Result 2 verbatim holds only in the well-mixed limit; a 3-site lattice test pinpoints where the real-algebraic formulation breaks down.

1 The counting problem is the bottleneck

Rare nucleation events in fluctuating environments—ice crystals in supersaturated air, amyloid fibrils at physiological monomer concentration, bacterial colonies on surfaces—involve discrete, countable entities in a continuous, fluctuating medium. Two field-theoretic traditions handle one half each. Doi–Peliti (DP) field theory Doi (1976); Peliti (1985) derives a path integral from a master equation; Martin–Siggia–Rose (MSR) Martin et al. (1973); Janssen (1976); De Dominicis (1976) derives one from a stochastic PDE. The coupled DP–MSR theory has recently been reviewed by Pruessner and Garcia-Millan Pruessner and Garcia-Millan (2025) and Bothe *et al.* Bothe et al. (2023), who demonstrated that coarse-grained density descriptions miss the particle structure and produce *spurious* entropy production; keeping the DP sector is essential.

A direct renormalisation-group programme for such coupled actions proceeds in five monolithic steps: (1) write the action; (2) enumerate Feynman diagrams; (3) evaluate loop integrals; (4) extract beta functions; (5) locate the instanton by complex-field saddle point. Steps (2)–(4) dominate the effort; step (5) is non-perturbative and usually approached separately. Changing the system (different reactions, different coupling) means redoing everything.

The point of this paper is that steps (2) and (5)—the *counting* of diagrams and the analogue for the instanton—can be factored out. Once factored, they reveal structure that is hard to see in the monolithic calculation. We call this factorisation **Coupled Field Analytic Combinatorics (CFAC)**. The CFAC decomposition is

$$\text{Observable} = \underbrace{\text{Counting}}_{\text{CFAC layer}} \times \underbrace{\text{Bridge}}_{\text{integral table}} \times \underbrace{\text{Ward correction}}_{\text{symmetry}}. \tag{1}$$

The CFAC layer enumerates 1PI graphs (finite), does power counting (mechanical), builds the Dyson–Schwinger equation (DSE) and extracts its singularity structure (algebraic). The bridge is a small pre-computed table of one-loop integrals indexed by topology and $(n_\partial, m_\psi/m_\phi)$, where n_∂ is the number of gradient insertions. The Ward layer is a finite set of symmetry checks.

The central recognition of this paper is that the Doi–Peliti generating function, as standardly constructed from a reaction network, *is already* a Lagrange equation of analytic combinatorics—once one knows to look for it. We state this as Theorem 1 (§2.5) after introducing the two technical ingredients, because the payoff is a body of imported mathematics: the transfer theorem for all-orders asymptotics, Puiseux classification of mean-field universality, and a new *branch-gap* contour integral for non-perturbative nucleation rates. The three demonstrations at coupled-field scope—Proposition 3 (one-loop perturbative exactness), Theorem 6 (the branch-gap formula), and Conjecture 8 (three-loop two-colour Kirchhoff factorisation)—are all consequences of applying Theorem 1 to a coupled DP–MSR system. The acronym CFAC (Coupled Field Analytic Combinatorics) reflects the ingredients: an algebraic formalism for coupled-field reaction-diffusion theories (Amarteifio, 2019) and the analytic combinatorics of Flajolet and Sedgewick (2009); §9 compares CFAC with the direct-RG approach of Pruessner and Garcia-Millan (2025).

2 Background: two imported tools

The approach of this paper combines two bodies of work that are largely foreign to the stochastic-field-theory mainstream and rarely appear in the Doi–Peliti/MSR literature Pruessner and Garcia-Millan (2025); Bothe et al. (2023); Täuber (2014). We introduce each separately and then describe the combination.

2.1 Algebraic reaction-diffusion field theory

The first ingredient is an algebraic formalism for stochastic reaction-diffusion systems developed in Amarteifio (2019), building on Doi–Peliti Doi (1976); Peliti (1985) and Heisenberg–Weyl Täuber (2014) machinery. The goal is to move from *deriving* field theories case-by-case (the standard approach) to *generating* them algorithmically from a chemical reaction network.

Reaction network as input. A network is specified by its stoichiometry matrix: for each reaction r, integers k_{ri}, l_{ri} giving the input and output counts of species A_i, together with a rate k_r.

The Doi shift and the Liouvillian. Each reaction $\sum_i k_{ri} A_i \to \sum_i l_{ri} A_i$ produces the Heisenberg–Weyl contribution

$$Q_r = k_r \Big[\prod_i (1+\tilde{\psi}_i)^{l_{ri}} - \prod_i (1+\tilde{\psi}_i)^{k_{ri}}\Big] \prod_i \psi_i^{k_{ri}}, \tag{2}$$

with $\psi_i, \tilde{\psi}_i$ creation/annihilation variables for species A_i (shifted by the Doi prescription Täuber (2014)). Expanding (2) gives a *vertex dictionary*: each monomial $g_{\{m_i\},\{n_i\}} \prod_i \tilde{\psi}_i^{m_i} \psi_i^{n_i}$ is a vertex with coupling g that depends algebraically on k_r and combinatorial prefactors.

From vertices to diagrammatic rules. Feynman rules follow mechanically: each vertex $\tilde{\psi}^m \psi^n$ has m outgoing and n incoming legs; propagators connect ψ to $\tilde{\psi}$; symmetry factors come from graph automorphisms. Spatial structure enters through the free propagator $G_0(k,\omega)^{-1} = -i\omega +$

$Dk^2 + m$, whose dimension fixes the upper critical dimension. Gradient insertions n_∂ come from spatial derivatives in interactions (drift, chemotaxis).

Kirchhoff polynomials from graphs. Every Feynman graph Γ with L loops, I internal edges, and V vertices has a Kirchhoff (first Symanzik) polynomial Bogner and Weinzierl (2010)

$$\mathcal{K}_\Gamma(\{u_e\}) = \sum_{T \in \mathrm{ST}(\Gamma)} \prod_{e \notin T} u_e, \tag{3}$$

the sum over spanning trees T of products of Schwinger parameters on edges *not* in T. By the matrix-tree theorem, $\mathcal{K}_\Gamma$ equals the determinant of any principal minor of the weighted graph Laplacian—an entirely algebraic construction.

What the algebraic formalism delivers. Given a reaction network, this formalism produces: the vertex dictionary; the 1PI Feynman graphs at any loop order; the symmetry factor and Kirchhoff polynomial of each; the upper critical dimension d_c; and the superficial degree of divergence

$$\omega(\Gamma, d) = dL - 2I + n_\partial^{\mathrm{tot}} \tag{4}$$

for every graph. *No loop integrals are evaluated.*

2.2 Analytic combinatorics (Flajolet–Sedgewick)

The second ingredient is analytic combinatorics (AC), a framework developed by Flajolet and Sedgewick (2009) for extracting asymptotic behaviour of combinatorial counts from the singularity structure of their generating functions. AC is standard in theoretical computer science and enumerative combinatorics (counts of trees, graphs, paths, partitions) but is not widely used in statistical physics, where the same asymptotic information is usually obtained from saddle points of actions, transfer matrices, or direct summation.

Why this should work for physics. The Doi–Peliti formulation already recasts a stochastic master equation as an operator algebra on a Fock space, with the Hamiltonian generating time evolution (Doi, 1976; Peliti, 1985; Täuber, 2014). In that language, observables are expectation values of monomials in $\tilde{\psi}, \psi$; Feynman diagrams compute them order by order; universality classes are read off from the relevant vertices near d_c. Analytic combinatorics sits one level up: the *generating function* that resums the Feynman series is a well-defined algebraic object—the solution of (5)—whose singularities encode the universality class directly. Three consequences make this useful for physics.

(i) Universality classes become polynomial invariants. The Puiseux exponent p/q at the DSE branch point classifies the mean-field behaviour: $p/q = 1/2$ is directed percolation and mean-field annihilation; $p/q = 1/3$ is the compact directed percolation / triple-root regime; higher-order branch mergers give further multi-critical points (Flajolet and Sedgewick, 2009; Amarteifio, 2019, 2026). Classification is algebraic, not dynamical.

(ii) The singularity is a handle on rare events. Because the transfer theorem (7) extracts both rate and prefactor from the same local data, it is naturally suited to questions about the asymptotic tails of the perturbation series—precisely the regime where rare-event and non-perturbative structure appear. Theorem 6 below makes this concrete: the nucleation barrier is a contour integral around the DSE branch point, in the same plane where the transfer theorem operates.

(iii) Graph-intrinsic substitution, no path integral needed. Reaction-diffusion on irregular substrates is notoriously awkward in the direct field-theoretic approach: one must either redo the RG with a non-standard propagator or appeal to scaling arguments. In AC the spectral dimension d_s enters through a single substitution in the engineering dimensions; no propagator is recomputed. This is the same trick that turns Weyl's law into a unified treatment of exponents on $\mathbb{Z}^d$, Sierpinski carpets, random trees, and scale-free networks (Amarteifio, 2026).

Identifying the AC constructions with stochastic-field-theoretic quantities is the bridge we exploit.

Dyson–Schwinger equation as Lagrange equation. For a scalar field theory with vertex function $\phi(G) = \sum_k g_k G^k$, the full propagator $G(z)$ with source-strength z satisfies the functional equation Bogner and Weinzierl (2010)

$$G(z) = z \cdot \phi(G(z)), \tag{5}$$

a closed implicit definition of G as an algebraic function of z. In AC language Flajolet and Sedgewick (2009), (5) is a *Lagrange equation*; the Lagrange inversion formula gives its Taylor coefficients in closed form.

Branch point and Puiseux exponent. The implicit function theorem applied to (5) fails at (G_*, z_*) satisfying $1 - z_*\phi'(G_*) = 0$ and $G_* = z_*\phi(G_*)$. This is the *branch point*. The Newton polygon of (5) near (G_*, z_*) determines the local Puiseux expansion

$$G(z) - G_* \sim C\,(z_* - z)^{p/q}, \qquad z \to z_*, \tag{6}$$

with rational exponent p/q. For a generic quadratic branch, $p/q = 1/2$; for a triple-root merger, $p/q = 1/3$.

Transfer theorem. The central *transfer theorem* Flajolet and Sedgewick (2009) converts local singularity behaviour into asymptotic coefficient growth:

$$[z^n]\,G(z) \sim \frac{C}{\Gamma(-p/q)}\,n^{-p/q-1}\,z_*^{-n}, \qquad n \to \infty. \tag{7}$$

Both the exponential rate z_*^{-n} *and* the algebraic prefactor are extracted from polynomial data alone. In the field-theoretic interpretation, $[z^n]G(z)$ is the n-th perturbative coefficient of the propagator; the transfer theorem is the rigorous all-orders statement of the perturbation series' singularity structure.

Spectral dimension. AC combines naturally with the spectral dimension Alexander and Orbach (1982); Havlin and Ben-Avraham (1987); Hattori et al. (1987) of an irregular substrate. On a graph with spectral dimension d_s, the return probability scales as $P(t) \sim t^{-d_s/2}$, and reaction-diffusion exponents follow from $d \to d_s$ in the engineering-dimension analysis—delivering exponents on fractals and cellular compartments without reworking the field theory.

2.3 Combining the two: CFAC

The two ingredients are complementary. Together they define what we call **Coupled Field Analytic Combinatorics (CFAC)**: a pipeline that takes a coupled-species reaction network and returns quantitative predictions from polynomial data. The algebraic formalism (§2.1) takes a chemical reaction network as input and produces Feynman graphs, Kirchhoff polynomials, and the DSE. Analytic combinatorics (§2.2) takes the DSE as input and produces asymptotic information: branch points, Puiseux exponents, and transfer-theorem asymptotics. The word *coupled* in CFAC refers both to the multi-species coupling (DP-sector particles interacting with MSR-sector fields) and to the *coupling* of the algebraic and AC layers into a single pipeline.

Combining them turns a reaction network directly into quantitative predictions, via the decomposition

$$\text{Observable} = \underbrace{\text{Counting}}_{\substack{\text{algebraic}\\ \text{formalism + AC}}} \times \underbrace{\text{Bridge}}_{\text{integral table}} \times \underbrace{\text{Ward correction}}_{\text{symmetry}}. \tag{8}$$

What AC gives that the algebraic formalism alone does not is the all-orders singularity structure of the perturbation series: finding the branch point of the DSE is equivalent to summing the full series asymptotically. What the algebraic formalism gives that AC alone does not is the DSE in the first place—and, crucially, the coupled DSE for a multi-species system, constructed from the stoichiometry matrix.

What analytical power does AC add over conventional pen-and-paper QFT? The honest answer is that AC does not reduce the intellectual work for any *single* computation; it reduces the marginal work for each new system and exposes structural connections that are difficult to see case-by-case. Five concrete additions:

1. *The DSE as a closed Lagrange equation.* Pen-and-paper writes $G = G_0 + G_0\Sigma G$ and treats Σ perturbatively; AC recognises $G = z\phi(G)$ as a Lagrange equation, from which $[z^n]G$ follows by closed-form Lagrange inversion rather than iterative loop summation.
2. *All-orders asymptotics via the transfer theorem.* Pen-and-paper gives the first few coefficients in a perturbative series by direct calculation. The transfer theorem (7) gives the full $[z^n]G$ asymptotic from a local calculation at the branch point, including the algebraic prefactor. This is the regime where Borel summation, resurgence, and non-perturbative structure enter.
3. *Polynomial-invariant classification of universality.* The Puiseux exponent p/q at the DSE branch point is a rational number determined by the polynomial structure. $p/q = 1/2$ is directed percolation; $p/q = 1/3$ is the compact-DP / triple-root class; higher-order mergers give further multi-critical classes. Classification becomes algebraic geometry on the DSE polynomial, not a case-by-case RG analysis.
4. *Exhaustive enumeration over "by inspection".* All-orders statements in pen-and-paper QFT usually require resummation, induction, or symmetry arguments. The algebraic formalism enumerates all 1PI graphs consistent with the vertex dictionary; power counting on that finite list converts an all-orders claim into a finite combinatorial check—as in Proposition 3.
5. *A common home for perturbative and non-perturbative physics.* Pen-and-paper derives perturbative exponents from loop integrals and instantons from action saddle points, using distinct machinery. AC places both at the branch point of the same generating function: the transfer theorem reads off perturbative asymptotics, and Theorem 6 reads off the non-perturbative instanton rate. The structural identification is new; the numerical values it produces at $d = 0$ are not.

For a reader familiar with pen-and-paper DP-MSR calculations, the tangible change is that repetitive machinery (diagram enumeration, power counting, singularity analysis) becomes mechanical, freeing analytical effort for the genuinely system-specific steps (identifying the vertex dictionary from the physics, evaluating the bridge integrals, and checking Ward identities).

2.4 Worked example: pair annihilation

For single-species pair annihilation $2A \to \varnothing$ at rate λ (stoichiometry k_A=2, l_A=0), the Liouvillian (2) reads

$$Q = \lambda\big[1 - (1+\tilde{\psi})^2\big]\,\psi^2 = -2\lambda\,\tilde{\psi}\psi^2 - \lambda\,\tilde{\psi}^2\psi^2, \tag{9}$$

after expanding $1-(1+\tilde{\psi})^2 = -2\tilde{\psi}-\tilde{\psi}^2$. The two resulting interaction vertices are $(m_A, n_A) = (1, 2)$ with coupling -2λ and $(2, 2)$ with coupling λ; we label them $g_{12} = -2\lambda$ and $g_{22} = \lambda$ (Doi, 1976; Peliti, 1985). A Pruessner-familiar reader will recognise these as the standard DP vertices for pair annihilation.

The mapping from these vertices to the DSE kernel $\phi(G)$ is the point where the algebraic formalism meets analytic combinatorics, and is worth spelling out. In the DSE (5), $G(z)$ is the full (dressed) particle propagator,[1] and z the bare propagator. Summing over all 1PI insertions, each interaction vertex $\tilde{\psi}^m\psi^n$ attaches to a vertex of G by contracting one $\tilde{\psi}$-leg (to the external line) and $(m-1)$ $\tilde{\psi}$-legs and n ψ-legs to internal dressed propagators—giving a contribution $g_{mn}\, G^{m+n-2}$. Summing all vertices plus the identity:

$$\phi(G) = 1 + \sum_{m,n:\, m+n\geq 3} g_{mn}\, G^{m+n-2}. \tag{10}$$

For pair annihilation: $\phi(G) = 1 + g_{12}G + g_{22}G^2 = 1 - 2\lambda G + \lambda G^2$. The DSE (5) is therefore $G = z[1 - 2\lambda G + \lambda G^2]$, a quadratic in G.

Solving: the branch point is at $G_* = 1/\lambda$, $z_* = 1/\lambda$; the Puiseux exponent is $1/2$. The transfer theorem (7) gives $[z^n]G(z) \sim n^{-3/2}\lambda^n$. Combined with Weyl's law on $\mathbb{Z}^d$ (substitution $d \to d_s$), this recovers the mean-field density decay $\rho(t) \sim t^{-d/2}$ for $d \geq 2$ and its breakdown for $d < 2$ (Täuber, 2014). A fully worked example for the Gribov/BRW process ($A \to 2A$, $2A \to A$, $A \to \varnothing$) is given in a companion note (Amarteifio, 2026). *No loop integral was evaluated in either case.*

2.5 Main recognition theorem

The preceding subsections separately developed the algebraic reaction-diffusion formalism (§2.1) and analytic combinatorics (§2.2). The central observation of this paper is that the object they share is already the same:

Theorem 1 (CFAC recognition)**.** *Let a reaction network have polynomial vertex structure (all vertices arising from the Doi–Peliti expansion of a finite stoichiometry matrix, projected if necessary to zero spatial dimensions). Then the Doi–Peliti dressed-propagator generating function $G(z)$, constructed from the Liouvillian of the network by the standard second-quantised field-theoretic route, satisfies*

$$G(z) = z \cdot \phi(G(z)), \qquad \phi(G) = 1 + \sum_{m+n\geq 3} g_{mn}\, G^{m+n-2}, \tag{11}$$

with ϕ a polynomial determined algorithmically by the stoichiometry. Equation (11) is a Lagrange equation in the sense of analytic combinatorics (Flajolet and Sedgewick, 2009). Consequently:

1. *All-orders coefficient asymptotics are given by the transfer theorem (7), with exponential rate z_*^{-n} and algebraic prefactor set by the Puiseux exponent at the branch point.*
2. *Mean-field universality classes are indexed by the Puiseux exponent p/q, a rational invariant of the polynomial ϕ at its branch point.*
3. *Non-perturbative nucleation rates are extracted by the branch-gap contour integral (20) of the same polynomial (Theorem 6 below).*
4. *Higher-loop UV divergences are classified by the superficial degree of divergence $\omega = dL - 2I + n_\partial^{\mathrm{tot}}$ evaluated on the finite enumeration of 1PI graphs consistent with the vertex dictionary.*

Why this is the whole paper. For a reader familiar with Doi–Peliti, steps (1)–(4) are not usually thought of as properties of a single object. They are handled by separate technical routes: loop integration and RG for perturbative exponents, action saddle-points for non-perturbative rates, case-by-case dimensional analysis for universality classes. Theorem 1 says they are all consequences of the *same* polynomial $\phi(G)$—the one that drops out of the algebraic formalism as soon as the Liouvillian is expanded. The physicist's DP generating function and Flajolet–Sedgewick's Lagrange equation are the same object, and the Flajolet–Sedgewick theorem apparatus transfers.

This is the clean content. The rest of the paper demonstrates the three non-trivial consequences at coupled-field scope: Proposition 3 is (4) applied to the coupled Keller–Segel sector; Theorem 6

[1] More precisely, $G(z)$ is the generating function $\sum_n n!\,\langle\psi^n\tilde{\psi}^n\rangle\, z^n$ of equal-time connected correlators in the DP path integral; z is the source strength. In AC language it is the *Lagrange generating function* for rooted 1PI trees (Flajolet and Sedgewick, 2009; Bogner and Weinzierl, 2010).

is (3) proved via change of variables; Conjecture 8 is a structural claim about how (4) behaves under bi-colouring of edges.

Scope. The theorem as stated holds when ϕ is polynomial in G alone. This covers pure reaction networks (DP) and the field-integrated projections of coupled theories used in Theorem 6. When gradient vertices carry momentum-dependent couplings (as in the spatially resolved coupled theory of §4), ϕ depends on momentum and the recognition becomes a functional statement; the transfer-theorem apparatus extends but is more involved. All coupled-field applications in this paper either project to $d = 0$ (nucleation, §10) or are single-loop perturbative (§4), so the polynomial form of Theorem 1 is sufficient.

3 Test case: coupled Keller–Segel

We take bacterial chemotaxis Keller and Segel (1970) as the vehicle. Discrete bacteria ψ diffuse (D_A) and are biased by an attractant field ϕ (diffusion D_c, decay κ). Bacteria source the field at rate μ; the gradient biases their motion with strength χ. The coupling is

$$S_{\text{int}} = -\chi \int \tilde{\psi}\,(\nabla\phi)\cdot(\nabla\psi) - \mu \int \tilde{\phi}\,\tilde{\psi}\,\psi. \tag{12}$$

The chemotaxis vertex has $n_\partial = 2$; the secretion vertex has $n_\partial = 0$. Saddle points reproduce the classical Keller–Segel PDE; linearising gives the mean-field instability threshold $n_c = D_A\kappa/(\chi\mu)$.

The direct-RG calculation for a related model was carried out by Gelimson and Golestanian Gelimson and Golestanian (2015); they integrated out the field first, obtaining a long-range Coulomb interaction, then applied RG to the single resulting field. We retain both fields explicitly.

4 CFAC result 1: one-loop perturbative exactness

4.1 The perturbative sector is pure bi-coloured graph theory

At the level of diagrams, a coupled DP–MSR theory is a field theory with two propagator types (DP and MSR, bi-coloured) and a vertex dictionary read off from the Liouvillian. Every perturbative observable—exponents, beta functions, fixed points—is determined by three algebraic data attached to each 1PI graph: the graph topology, its Kirchhoff polynomial $\mathcal{K}_\Gamma(\{u_e\})$ with edges coloured DP or MSR, and the Symanzik polynomial containing the propagator masses m_ψ, m_ϕ and diffusion constants D_A, D_c.

Proposition 2 (Perturbative coupled-field theorem)**.** *For any coupled DP–MSR reaction network with polynomial vertex structure, the full perturbative renormalisation-group flow is determined by the bi-coloured Kirchhoff and Symanzik polynomials on the enumerated 1PI graphs. Specifically:*

1. *1PI graphs at loop order L are enumerated by the algebraic formalism from the vertex dictionary (finite list per L).*
2. *Each graph has a two-colour Kirchhoff polynomial $\mathcal{K}_\Gamma(p, f)$ (matrix-tree theorem on the weighted Laplacian with edge colours $p = G_\psi$, $f = G_\phi$).*
3. *The superficial degree of divergence $\omega(\Gamma, d) = dL - 2I + n_\partial^{\text{tot}}$ classifies which graphs are UV divergent; the finite list of divergent graphs determines the beta functions at each loop order.*
4. *Mass ratios (m_ψ/m_ϕ) and diffusion ratios (D_A/D_c) enter only through the Symanzik polynomial evaluated at the saddle; bi-coloured factorisation (Conjecture 8) splits these dependencies into single-colour factors.*

Significance. The proposition generalises Result 1 below from the KS-specific coupling to an arbitrary coupled DP–MSR reaction network with polynomial vertices. Every such system is handled by the same algebraic pipeline; only the Kirchhoff- and Symanzik-polynomial inputs change. KS, a field-coupled Schlögl model, coupled catalysis networks, multi-species chemotaxis—the *perturbative* RG flow of all of these is determined by the same construction, with no case-by-case derivation of Feynman rules. The one-loop exactness of the minimal KS coupling sector (Corollary 3 below) is a specific instance in which the vertex content is so constrained that the divergent graph list is finite even at all orders.

4.2 Corollary: one-loop exactness of the minimal KS coupling sector

Proposition 3 (One-loop perturbative exactness of the coupling sector)**.** *Consider the minimal coupled theory defined by the two coupling vertices in* (12) *together with free DP and MSR propagators and* no additional interaction vertices *(i.e. no DP branching/death or MSR noise vertices). The bacterial self-energy of this theory has a single UV divergence at one loop,*

$$\delta D_A^{(1)} = +\frac{\chi\mu\, D_c D_A}{\pi(D_A + D_c)^3}\,\frac{1}{\varepsilon} + O(\varepsilon^0), \quad \varepsilon = 2 - d, \tag{13}$$

and no *UV divergences at any loop order $L \geq 2$. All vertex corrections and the attractant self-energy are UV finite; $Z_\chi = Z_\mu = Z_\phi = 1$; the beta function of $g = \chi\mu/[2\pi(D_A + D_c)^2]$ is*

$$\beta_g = -\varepsilon g - \frac{4D_A D_c}{(D_A + D_c)^2}\, g^2 \tag{14}$$

exactly *to all perturbative orders in this minimal sector.*

Scope. The proposition is a statement about the *coupling sector* of the KS theory only. Adding DP branching ($\tilde{\psi}^2\psi$, which governs the directed-percolation universality class Täuber (2014)) or DP death ($\tilde{\psi}\psi$ mass term) introduces additional vertices that enumerate to new divergent graphs at higher loop orders; the all-orders $\omega_L < 0$ bound derived below no longer holds uniformly. Adding MSR noise ($\tilde{\phi}^2$) modifies the attractant self-energy. Result 1 therefore isolates the perturbative behaviour of the *coupling alone*, disentangled from the well-studied DP and MSR divergences.

What CFAC does, and what a human/code does at each step. The proposition claims an *all-orders* statement: that no UV divergence exists beyond one loop. In the direct-RG route, this would require computing a potentially infinite sequence of integrals; CFAC reduces it to a finite, mechanical inspection. Concretely, the pipeline is:

1. *Input (human):* the vertex dictionary from (12)—two coupling vertices χ (3-leg, $n_\partial = 2$) and μ (3-leg, $n_\partial = 0$), plus the free DP and MSR sectors.
2. *Counting L (algebra):* for 3-leg vertices and $E = 2$ external legs, leg-counting $3V = 2I + E$ plus the Euler relation $L = I - V + 1$ give $V = 2L$, $I = 2L + 1$ at loop order L. Two lines of algebra.
3. *Graph enumeration (code):* enumerate all non-isomorphic 1PI graphs with these (V, I, E) and compatible leg types. At $L = 2$: the unique topology is $K_4 \setminus e$; at $L = 3$: the unique topology is the triangular prism (after automorphism). Vertex-consistent edge colourings are enumerated directly (`paper/reproduce.py`, ~ 10 lines of `sympy`).
4. *Power counting (algebra):* apply $\omega = 2L - 2I + n_\partial^{\text{tot}}$ to each coloured graph. One subtraction per diagram.
5. *Verdict (by inspection):* every diagram at $L \geq 2$ has $\omega < 0$, so every diagram at $L \geq 2$ is UV convergent.

What ω is doing. The superficial degree of divergence $\omega = dL - 2I + n_\partial^{\mathrm{tot}}$ is standard QFT power counting (Täuber, 2014): a loop diagram in d dimensions has dL momentum-integration powers, $-2I$ from propagator falloff at large k, and $+n_\partial^{\mathrm{tot}}$ from gradient insertions. $\omega \geq 0$ means UV divergent (contributes to renormalisation); $\omega < 0$ means UV finite. We apply this formula to each enumerated 1PI graph.

At $d = 2$ the finite counts are:

L	graphs	vertex content	propagators	$n_\partial^{\mathrm{tot}}$	ω at d=2
1	1	$\chi\,\mu$	$G_\psi + G_\phi$	2	0 (log)
2	6	$\chi^2\mu^2$	$3G_\psi + 2G_\phi$	4	-2
3	20	$\chi^3\mu^3$	$5G_\psi + 3G_\phi$	6	-4

The pattern continues for every $L \geq 2$: because the coupled KS vertex content is fixed by the reaction network (every self-energy graph at order L has L copies of χ and L of μ, giving $n_\partial^{\mathrm{tot}} = 2L$ and $I = 2L + 1$), substitution into ω at $d = 2$ yields

$$\omega_L = 2L - 2(2L+1) + 2L = -2 \quad \text{for every } L \geq 1. \tag{15}$$

At $L = 1$ this is corrected by a log divergence from the gradient structure of χ giving $\omega_1 = 0$; at $L \geq 2$ no such correction exists and every graph is UV finite.

What is *not* done: no loop integral is evaluated, no action is saddle-pointed, no beta function is re-derived. The all-orders claim reduces to the single algebraic inequality (15). The existing formulation in the literature (Gelimson and Golestanian, 2015; Pruessner and Garcia-Millan, 2025) does not organise the computation this way: it computes the one-loop integral explicitly (as we do in the bridge step below), but has no mechanism to rule out higher-loop divergences without further integral evaluation.

4.3 A dictionary-level criterion for one-loop UV-exactness

Proposition 3 establishes one-loop exactness of the minimal KS coupling sector by exhaustive enumeration at $L \leq 3$ plus the all-orders bound (15). That argument lifts to a dictionary-level sufficient condition: given *any* Doi–Peliti vertex dictionary, a single inequality decides whether UV divergences can appear above one loop.

Setup. Let $\mathcal{V} = \{v_1, \ldots, v_N\}$ be a polynomial vertex dictionary, each vertex a monomial in $\tilde\psi, \psi, \tilde\phi, \phi$ with total leg count $k_v = m_v + n_v \geq 3$ and gradient count $n_\partial(v) \in \{0, 2, 4, \ldots\}$. Define the *divergence weight*

$$\tau_v \;:=\; \frac{n_\partial(v) - 2}{k_v - 2}, \qquad \tau_*(\mathcal{V}) \;:=\; \max_{v\in\mathcal{V}} \tau_v. \tag{16}$$

Proposition 4 (Doi–Peliti one-loop UV-exactness criterion)**.** *Fix a two-point 1PI amplitude ($E = 2$). Then uniformly over 1PI topologies at loop order L,*

$$\omega_{\max}(L; d) \;=\; [(d-2) + 2\,\tau_*]\,L \;+\; 2. \tag{17}$$

Consequently:

1. (Sufficiency) *If $d \leq 2 - 2\tau_*$, every 1PI self-energy graph with $L \geq 2$ has $\omega < 0$, so the theory is UV divergent at most at $L = 1$.*
2. (Upper critical dimension) $\omega_{\max}(1, d_c) = 0$ *at $d_c = -2\tau_* = \min_v 2(2 - n_\partial(v))/(k_v - 2)$.*
3. (Tightness) (17) *is attained by the graph concentrated on the vertex type achieving τ_*, provided that vertex admits a 1PI self-energy topology under the Doi–Peliti leg-balance constraint (Remark 5).*

Proof. Euler's relation $L = I - V + 1$ and leg counting $2I + E = \sum_v k_v$ with $E = 2$ give $\omega = (d-2)L - 2V + 2 + n_\partial^{\rm tot}$. Writing $V = \sum_k V_k$ and $\sum_k (k-2)V_k = 2L$, a grouping gives $\omega \le (d-2)L + 2 + \sum_k \tau_k (k-2)V_k$ with $\tau_k = [\bar{n}_\partial(k) - 2]/(k-2)$. Maximising subject to $\sum_k (k-2)V_k = 2L$ gives $2L\tau_*$, proving (17). Tightness is by single-vertex saturation (modulo Remark 5); items (1)–(2) follow by inspection. □

Remark 5 (Field-type balance)**.** *The bound* (17) *ignores the Doi–Peliti asymmetry between* $\tilde\psi, \tilde\phi$ *and* ψ, ϕ *legs: internal propagators pair a tilde-leg to a non-tilde-leg of the same species, constraining* $\sum_v m_v^\psi = \sum_v n_v^\psi$ *and similarly for* ϕ*. A single-vertex-type saturation is legal only when the maximising vertex is* self-pairing*. When it is not (as in the KS sector, where* χ *and* μ *must appear in equal numbers), the sharp bound is a linear fractional program over the field-balanced cone, solvable in closed form by Charnes–Cooper for dictionaries with* $\lesssim 4$ *vertex types. For the KS sector the coarse bound gives* $d_c = 0$ *(loose); the balanced LP recovers the sharp* $d_c = 2$ *consistent with* (15).

Theory	vertices (k, n_∂)	τ_*	d_c (coarse)	d_c (balanced)	lit. d_c
Pure DP ($A \leftrightarrow 2A$)	$(3,0)$	-2	4	4	4 (Täuber, 2014)
Pair annihilation $2A \to \varnothing$	$(4,0), (3,0)$	-1	2	2	2
KS coupling sector (Result 1)	$(3,2), (3,0)$	0	0	2	2
ϕ^4 scalar (non-CRN)	$(4,0)$	-1	2	2	$4^\dagger$

†The ϕ^4 entry is not a failure of the criterion: at its UCD $d = 4$, ϕ^4 has quadratic and logarithmic primitive divergences at *every* loop order and is not one-loop UV-exact; the criterion correctly does not flag $d = 4$ as exact. Below the UCD (e.g. $d = 2$) it is super-renormalisable and the criterion correctly reports one-loop exactness.

Scope. Prop. 4 controls positive-power UV divergences: it is sufficient for the universality claim (critical exponents from the one-loop β-function) but does not rule out sub-leading logarithmic contributions to higher-loop beta coefficients. The stronger "exact exponents to all orders" statement of Proposition 3 requires in addition that the one-loop β-function close on itself, verified separately via (14).

Commentary. The criterion converts the construction-specific all-orders power-counting of Result 1 into a mechanical inspection of the vertex dictionary: compute τ_v for each vertex, take the max, compare with the spatial dimension. Any coupled reaction-field theory can be classified before any Feynman integral is written. The minimal KS coupling sector is in this light a *discovery* of the criterion, not a coincidence: it is the smallest non-trivial two-species dictionary with $\tau_* \le 0$, forced to be one-loop UV-exact in $d \le 2$ by the vertex algebra alone. Adding a DP branching vertex $\tilde\psi^2\psi$ ($\tau_v = -2$) preserves the exactness at $d \le 2$; adding a gradient-laden quartic raises τ_* and destroys it. This is the precise content of "CFAC does the power counting for you."

The one-loop integral. The bridge evaluates (13) once. Direct quadrature of the 2D integral at cutoffs $\Lambda \in \{10, 20, 50, 100, 200\}$ matches (13) to five significant figures (ratio = 1.0000 at $\Lambda = 100, 200$). The positive sign means stochastic field fluctuations *stabilise* the system against collapse, raising the critical density above its mean-field value. The exact dynamic exponent is $z = 2 - \varepsilon$; quantitative extrapolation to $\varepsilon = 1$ is unreliable, as standard for one-loop results Täuber (2014).

Direct numerical confirmation in $d = 2$. Beyond the integral check, we simulated the full stochastic KS model on a 2D periodic lattice and extracted the effective diffusivity at two scales by fitting $\langle (\Delta x)^2 \rangle \sim D_{\rm eff} t^\alpha$ separately at early and late times. One-loop theory (13) predicts $D_{\rm late} > D_{\rm early}$, and the excess growing with χ. We observe exactly this, with $\alpha \approx 0.98$ across the whole range:

χ	D_{early}	D_{late}	$D_{\text{late}}/D_{\text{early}}$	α
0.0	0.955	1.015	1.0622	0.980
0.3	0.952	1.011	1.0619	0.980
0.5	0.950	1.009	1.0618	0.980
1.0	0.946	1.004	1.0612	0.980
2.0	0.936	0.992	1.0604	0.980

The ratio exceeds unity and the large-scale diffusivity is raised by $\sim 6\%$, consistent in sign with Proposition 3.

Comparison. The direct-RG approach to the related model in Gelimson and Golestanian (2015) obtains a beta function from the one-loop Coulomb interaction and finds a stable fixed point below the collapse threshold (consistent with stabilisation). The AC approach additionally establishes, by enumeration, that higher loops contribute nothing to the UV divergence structure—a claim that is independent of the parameterisation (field integrated out vs. retained).

5 CFAC result 2: branch-gap instanton formula

Background: what is an instanton? An *instanton* is a classical solution that dominates the path integral representation of a rare event—typically the optimal fluctuation carrying a system from one metastable state to another over a barrier Langer (1967); Täuber (2014). For a stochastic process with action $S[\mathbf{x}(t)]$, the probability of a rare transition is $P_{\text{rare}} \sim \exp(-S_{\text{inst}})$, where S_{inst} is the action evaluated on the extremising trajectory. In a many-body system with system size V, this gives rate $\tau^{-1} \sim \exp(-V \cdot S)$ with S the *intensive* barrier. Computing S generally requires finding a saddle point of the action in complex field space—a non-perturbative calculation separate from the Feynman-diagram expansion.

For a birth-death process with rates $w^+(x), w^-(x)$ (continuum limit $x = n/V$), the WKB treatment Elgart and Kamenev (2004); Assaf and Meerson (2017); Dykman et al. (1994); Langer (1967) expresses the nucleation barrier as the quasi-potential integral

$$S_{\text{WKB}} = \int_{x_1}^{x_2} \ln\bigl[w^-(y)/w^+(y)\bigr]\,\mathrm{d}y, \tag{18}$$

between the metastable fixed point x_1 and the unstable barrier x_2 (both roots of $w^+ = w^-$). The corresponding mean first-passage time scales as $\tau \sim C\exp(VS)$ with a prefactor C set by fluctuations around the instanton path. Equation (18) is the standard x-space form.

We show that this same action equals a *branch-gap integral* of the DSE generating function—an integral in z-space around a polynomial singularity. For a coupled system with $w^+(x) = a + \lambda x^2$, $w^-(x) = \delta x + dx^3$ (obtained by integrating out the field at steady state, so $\lambda = g\mu/\kappa$), the DSE equation $w^-(x(z)) = z\,w^+(x(z))$ reads

$$d\,x^3 - \lambda z\,x^2 + \delta\,x - az = 0. \tag{19}$$

Theorem 6 (Branch-gap identity)**.** *With z_* the branch point of* (19) *(a real root of its discriminant) and $x_{\text{unst}}(z)$, $x_{\text{stab}}(z)$ the two real branches that merge at z_*, the nucleation barrier is*

$$S = \int_1^{z_*} \frac{x_{\text{unst}}(z) - x_{\text{stab}}(z)}{z}\,\mathrm{d}z \;=\; S_{\text{WKB}}. \tag{20}$$

Sketch. Set $z(y) = w^-(y)/w^+(y)$. Both endpoints x_1, x_2 map to $z = 1$ (fixed points have $w^+ = w^-$). The integration path from x_1 (stable, $z = 1$) to x_2 (unstable, $z = 1$) passes through the branch point z_* while switching from x_{stab} to x_{unst}. Change of variables and integration by parts; boundary terms vanish at $z = 1$ and cancel at z_*; the result is (20). Appendix B. □

Is this new? The x-space form (18) is classical (Freidlin–Wentzell theory, Onsager–Machlup, WKB for population dynamics Elgart and Kamenev (2004); Assaf and Meerson (2017); Dykman et al. (1994)). The z-space form (20), and its identification of the instanton with a contour integral around a polynomial branch point, appear to be new. The two integrals have the same value but different structure.

Honest accounting. It is worth being precise about what the reformulation does and does not buy. (i) The cubic (19) has closed-form roots (Cardano), so $x_{\text{stab}}(z)$ and $x_{\text{unst}}(z)$ are algebraic functions of z—the "no logarithm" phrasing is colour: the integral $\int \text{gap}/z\, \mathrm{d}z$ is still not elementary, and we evaluate it numerically. (ii) The x-space form integrates along the deterministic trajectory from x_1 to x_2; the z-space form integrates along a contour from $z = 1$ to z_*. Different parametrisation of essentially the same geometric object, not a structural shortcut in integration cost at the $d = 0$ level. (iii) What the z-space form *does* buy is that the branch point z_* is a discriminant zero of an explicit polynomial, so the full identification requires no deterministic simulation, no Langevin trajectory, and no evaluation of $w^{\pm}(x)$ at intermediate points. For systems where the reaction network is known but the deterministic trajectory is not in closed form, or for higher-degree resultants (multi-species coupled systems, where Cardano is replaced by the generic resultant of two polynomials), this becomes a real gain. (iv) Crucially, the branch-gap formulation exposes the instanton as a feature of the *same* generating-function singularity that the transfer theorem uses to sum the perturbative series—a structural connection not visible in the x-space form. (v) The theorem here is for $d = 0$ (well-mixed); the spatial generalisation requires replacing the branch-gap integral by a functional determinant over critical droplets, and is not attempted here.

Why AC makes this possible. The DSE generating function is the natural object of AC. Its branch point is the singularity that the transfer theorem extracts to get asymptotic coefficients. That the same object gives the non-perturbative instanton is a direct bridge between perturbative (AC) and non-perturbative (WKB) structure. In the direct-RG approach one computes the instanton separately as a saddle point of the action in complex field space; in AC the same number falls out of the polynomial structure of the DSE.

Verification. Seven parameter sets spanning $a \in [0.02, 0.5]$, $\lambda \in [2.5, 6.0]$, $\delta \in [2.0, 5.0]$, $d \in [0.5, 2.0]$; ratio $S_{\text{gap}}/S_{\text{WKB}} = 1.000000$ to six significant figures in all cases (Appendix B). One set has $z_* \approx 7.5$ (large branch point, outside naive search); widening the scan recovers the identity. Gillespie simulations at $V = 5$–15 confirm $\tau_{\text{MFPT}} \approx C \exp(VS)$ with the S-values (20).

6 CFAC result 3: two-colour Kirchhoff via Laplacian pencil

The Kirchhoff polynomial $\mathcal{K}(G; \{u_e\})$ controls the parametric integral over Schwinger parameters Bogner and Weinzierl (2010). For a Feynman graph carrying two edge colours—P for G_ψ-propagators and F for G_ϕ-propagators—specialise $u_e = p$ on colour-P edges and $u_e = f$ on colour-F edges. The bivariate specialisation $\mathcal{K}(G; p, f)$ is what governs whether the particle and field sectors decouple at the level of Schwinger integrals.

6.1 Matrix-tree pencil theorem

Let $\widetilde{L}_P(G)$ and $\widetilde{L}_F(G)$ be the reduced graph Laplacians of the spanning subgraphs $G \restriction E_P$ and $G \restriction E_F$ obtained by keeping only P- or F-edges respectively (with the full vertex set). Kirchhoff's matrix-tree theorem together with the bilinear structure of the weighted Laplacian gives the following unconditional identity.

Theorem 7 (Two-colour Kirchhoff pencil)**.** *For any connected multigraph G with two-colouring $c : E \to \{P, F\}$,*

$$\mathcal{K}(G; p, f) \;=\; \det(p\,\widetilde{L}_P(G) + f\,\widetilde{L}_F(G)), \tag{21}$$

a homogeneous binary form of degree $n-1$ in (p, f), where $n = |V(G)|$. When $\widetilde{L}_P$ is invertible, this factors over $\mathbb{R}$ as

$$\mathcal{K}(G; p, f) \;=\; \det(\widetilde{L}_P) \prod_{i=1}^{n-1} (p + \mu_i f), \tag{22}$$

with $\mu_1, \ldots, \mu_{n-1}$ the generalised eigenvalues of the Laplacian pencil $(\widetilde{L}_F, \widetilde{L}_P)$ (solutions of $\det(\widetilde{L}_F - \mu\widetilde{L}_P) = 0$). Both reduced Laplacians are symmetric and positive semidefinite, so every $\mu_i \geq 0$.

Proof. The weighted graph Laplacian is linear in edge weights: substituting $u_e = p$ (resp. f) gives $\widetilde{L}(G; p, f) = p\widetilde{L}_P + f\widetilde{L}_F$. The matrix-tree theorem gives (21). When $\widetilde{L}_P$ is invertible, $\det(p\widetilde{L}_P + f\widetilde{L}_F) = \det(\widetilde{L}_P) \det(pI + f\,\widetilde{L}_P^{-1}\widetilde{L}_F)$; the second determinant is the reversed characteristic polynomial of $\widetilde{L}_P^{-1}\widetilde{L}_F$, similar to the symmetric matrix $\widetilde{L}_P^{-1/2}\widetilde{L}_F\widetilde{L}_P^{-1/2}$ which has non-negative real spectrum. □

Theorem 7 replaces the earlier formulation of this result as a conjecture: the $\mathbb{R}$-splitting of $\mathcal{K}(G; p, f)$ into $n-1$ linear factors is a *theorem*, not a conjecture, for every 1PI Feynman graph and every colouring.

Physical consequence (unconditional). The Schwinger integrand $\mathcal{K}(G; p, f)^{-d/2}$ therefore decomposes over $\mathbb{R}$ as $\det(\widetilde{L}_P)^{-d/2} \prod_i (p + \mu_i f)^{-d/2}$: after a linear change of Schwinger parameters that diagonalises the pencil, the bridge integral becomes a product of $n-1$ independent one-variable integrals. The "bridge table grows combinatorially at each loop order" worry is thereby dissolved unconditionally: the combinatorial explosion is absorbed into a *spectral* computation of the pencil $(\widetilde{L}_F, \widetilde{L}_P)$, linear in n.

6.2 What factorises over $\mathbb{Q}$: refined conjecture

The question originally posed as "Conjecture 3" is the arithmetic refinement: *are the pencil eigenvalues μ_i rational?* An affirmative answer gives a factorisation of $\mathcal{K}(G; p, f)$ as a product of $n-1$ rational linear forms over $\mathbb{Z}[p, f]$; a fully rational pencil is the strong form of the old conjecture. The naive polynomial identity $\mathcal{K}(G; x) = \mathcal{K}_P(G|_P)\,\mathcal{K}_F(G|_F) \cdot C(x)$ in the full edge-variable ring, however, is *false already at $L = 2$*: the theta graph Θ_3 with colouring (P, P, F) gives $\mathcal{K} = x_1 + x_2 + x_3$ while $\mathcal{K}_P \cdot \mathcal{K}_F = (x_1 + x_2)x_3$, and the ratio is not a polynomial. The refined, correct statement is bivariate.

Conjecture 8 (Rational-pencil conjecture, refined)**.** *For every vertex-consistent 2-colouring of every 1PI KS self-energy graph, the Laplacian pencil $(\widetilde{L}_F, \widetilde{L}_P)$ has at least one rational eigenvalue; equivalently, $\mathcal{K}(G; p, f)$ has at least one $\mathbb{Q}$-linear factor.*

A sufficient condition is the following *balance lemma.*

Lemma 9 (Colour-balance implies rational factor)**.** *If there exist integers $(\alpha, \beta) \neq (0, 0)$ and a non-constant vertex function $v : V \to \mathbb{Z}$ with $(\alpha L_P + \beta L_F)\,v = 0$, then $(\alpha p + \beta f)$ divides $\mathcal{K}(G; p, f)$ in $\mathbb{Z}[p, f]$.*

Proof sketch. The matrix $\alpha L_P + \beta L_F$ has kernel of dimension at least two ($\mathbf{1}$ and v), so any reduced version is singular, i.e. $\mathcal{K}(G; \alpha, \beta) = 0$. Hence $(\alpha p + \beta f)$ is a linear factor of the homogeneous $\mathcal{K}(G; p, f)$; integrality follows by Gauss's lemma. □ □

The $L = 3$ empirical result of the CFAC paper—254 of 256 colourings factorise, all 182 physically relevant colourings factorise—is consistent with Conjecture 8: each such colouring furnishes a balance vertex function. The remaining pencil eigenvalues may or may not be rational, but Theorem 7 already gives the decoupling-over-$\mathbb{R}$ consequence.

Obstruction to the strong form. For graphs with $n-1 \geq 3$, the pencil characteristic polynomial is a cubic or higher with no *a priori* rational root; generic KS-consistent colourings of $L \geq 4$ topologies (e.g. prism $K_3 \square K_2$, with degree-5 characteristic polynomial) can be expected to produce $\mathbb{Q}$-irreducible factors. The strong "all $\mu_i \in \mathbb{Q}$" statement is therefore unlikely to hold; the refined Conjecture 8 (at least one rational μ_i) is the version supported by structural evidence.

What this means physically. For CFAC the unconditional pencil factorisation (22) is already the payoff: the bridge table's size is governed by the number of distinct pencil spectra, not the number of edge colourings, and each spectrum yields a product of $n-1$ independent single-integral bridge entries. The two colour sectors need not mix irreducibly at the level of Schwinger integrals. The rationality question (Conjecture 8) refines this by asking whether those bridge entries admit closed-form rational arithmetic rather than numerical spectrum, but is not required for the factorisation to hold as an integration-theoretic statement.

7 Spectral substitution for coupled systems

Reaction-diffusion exponents on an irregular substrate follow from $d \to d_s$ Alexander and Orbach (1982); Havlin and Ben-Avraham (1987); Hattori et al. (1987). For coupled systems with $D_A \neq D_c$, one could worry that different fields see different effective substrates.

Conjecture 10 (Universality of d_s)**.** *Both fields see the same d_s; the substitution applies uniformly to both sectors.*

Sierpinski gasket at level 6 (1095 vertices, exact $d_s = 2\ln 3/\ln 5 = 1.365$). Return probability $P(t) \sim t^{-d_s/2}$ from 8000 random walks at hopping probability $D \in \{0.1, 0.3, 0.5, 1.0\}$:

D	0.1	0.3	0.5	1.0	exact
d_s	1.386	1.361	1.352	1.388	1.365

All within 2% of the exact value and independent of D, as expected for a graph-intrinsic property. (Finite-size effects are significant: the same measurement at level 4, with only 123 vertices, gives apparent $d_s \in \{0.14, 0.49\}$—recovery of d_s requires a graph large enough that the random walk samples multiple hierarchical levels.)

8 Tropical bridge: the one-loop bubble is algebraically exact

A natural question is whether the *bridge table* of loop integrals (§4) can itself be expressed in closed algebraic form, rather than left as numerical quadratures. The answer at one loop is affirmative, and the relevant technology is *tropical geometry* Borinsky (2020); Borinsky et al. (2025); Arkani-Hamed et al. (2022): the integrand is written in Schwinger form, the first Symanzik polynomial F is viewed as a tropical polynomial on the projective simplex of Schwinger parameters, and its Newton polytope determines the asymptotic structure.

For the one-loop DP self-energy bubble in $d = 2$,

$$I_{\mathrm{DP}}(p; m_1, m_2) = \int_0^1 \mathrm{d}t\, \frac{1}{F(t; p, m_1, m_2)}, \quad F = m_1^2 t^2 + (p^2 + m_1^2 + m_2^2)\, t(1-t) + m_2^2 (1-t)^2,$$

the Newton polytope is the 1-simplex with vertex monomials at $t \to \{0, 1\}$ and an edge-interior monomial at $t = 1/2$. Applying the Arkani-Hamed–Hillman–Salvatori "leading-facet plus Hessian" prescription Arkani-Hamed et al. (2022) yields the closed form

$$\boxed{I_{\mathrm{DP}} = \frac{1}{\sqrt{\lambda_E}} \ln\left[\frac{\left(\sqrt{\lambda_E} + p^2 + m_1^2 + m_2^2\right)^2}{4\, m_1^2\, m_2^2}\right]}, \tag{23}$$

where the Euclidean Källén-like discriminant is

$$\lambda_E(p^2, m_1^2, m_2^2) = (p^2 + m_1^2 - m_2^2)^2 + 4p^2 m_2^2 = \det(\text{tropical Hessian at saddle}).$$

The prefactor $1/\sqrt{\lambda_E}$ is the Gaussian fluctuation determinant around the tropical saddle point; the logarithm is the sum of Newton-vertex boundary contributions.

Proposition 11 (tropical one-loop exactness)**.** *For any one-loop two-propagator diagram in the coupled DP–MSR theory, the AHHS tropical-saddle + Hessian formula* (23) *evaluates the Schwinger integral* exactly*, not asymptotically.*

Proof sketch. The Symanzik polynomial F is homogeneous of degree $L + 1 = 2$ in the Schwinger parameters; on the projective simplex it is a non-degenerate quadratic form whose Hessian is constant. A non-degenerate Gaussian integral against $F^{-(d/2)}$ is evaluated exactly by its saddle plus Hessian prefactor; no higher corrections arise because the Taylor expansion of $\ln F$ around the saddle terminates at second order up to a total derivative. The log term collects the boundary contributions from the two Newton vertices. □ □

Numerical verification across 14 orders of magnitude of kinematic ratio p^2/m^2 and m_1^2/m_2^2 confirms (23) to machine precision. By contrast, the crude single-vertex tropical approximation $I \sim 1/F_{\min}$ is off by factors 10^2–10^7 in non-symmetric regimes, and a three-cell Newton-polytope dissection without the Hessian overshoots by 10–70%; the Hessian is essential.

Consequences for the CFAC bridge. (a) The coupling-sector one-loop table is not a numerical dictionary but a closed algebraic expression parameterised by (p^2, m_1^2, m_2^2). (b) Branch cuts (the Minkowski threshold $s = (m_1 + m_2)^2$) appear as the vanishing locus $\lambda_E = 0$ of the Hessian determinant—directly visible on the Newton polytope. (c) For higher-loop diagrams the Symanzik polynomial has degree $L + 1$ and is no longer quadratic; genuine multi-facet sums are required, for which Borinsky's tropical Monte Carlo Borinsky (2020); Borinsky et al. (2025) gives an exact integration procedure. At one loop—the setting of Proposition 3 (perturbative exactness in the minimal KS coupling sector)—the tropical form is exact and CFAC is purely algebraic from reaction network through fixed-point exponent.

When one loop is the whole story. For CRNs where the superficial-degree enumeration of (C4) establishes *one-loop exactness of the UV divergence structure*—i.e. $\omega(L) < 0$ for every topology with $L \geq 2$, as we prove in Proposition 3 for the minimal KS coupling—Corollary **??** of the companion theorem (**?**) closes the *entire* perturbative pipeline in closed algebraic form: reaction network → vertex dictionary → Lagrange equation → Puiseux exponent + branch-gap + tropical bridge, with no numerical quadrature anywhere. This is not a narrow coincidence: the standard coupled-field literature (pair annihilation, pure DP, BARW, Gelimson–Golestanian chemotaxis, Pruessner–Garcia-Millan KS) stops at one loop precisely because the -enumeration of those systems rules out higher-loop UV divergences. Where CFAC genuinely fails to close in symbolic form is the residual class of CRNs for which the enumeration does not yield one-loop exactness (e.g. quartic ψ^4 active-birth sectors with persistent two-loop divergences); there one keeps tropical as an algorithmic rather than symbolic tool via Borinsky Borinsky (2020); Borinsky et al. (2025).

9 Comparison with direct-RG programmes

The direct-RG approach to coupled particle-field theories Pruessner and Garcia-Millan (2025); Bothe et al. (2023); Gelimson and Golestanian (2015) typically proceeds as follows. The coupled action is written explicitly; for active matter and chemotactic systems the field is often integrated out to produce a single-field action with nonlocal (e.g. Coulomb) interaction; Feynman rules are derived; one-loop integrals are computed explicitly; fixed points and exponents are extracted from the resulting beta function.

The AC approach differs in emphasis. Side-by-side:

Task	**Direct RG (e.g. Pruessner and Garcia-Millan (2025); Gelimson and Golestanian (2015))**	**CFAC (this work)**
Enumerate diagrams	by hand or from Feynman rules, system-specific	from reaction network, automatic
Power counting	from field dimensions at each vertex	$\omega = dL - 2I + n_\partial^{\text{tot}}$ on enumerated graphs
Loop integrals	computed per system	once, tabulated (bridge)
Instanton (nucleation rate)	complex-field saddle point, system-specific	branch-gap integral of the DSE polynomial, Theorem 6
All-orders claims	require resummation or explicit higher-loop work	direct from graph enumeration (Prop. 3)
Change the system	redo everything	change the vertex dictionary; rest automatic

A reader familiar with the Pruessner–Garcia-Millan or Gelimson–Golestanian calculations can cross-check this paper as follows. Equation (13) is the one-loop bacterial self-energy computed in the two-field formalism; it reduces to the Coulomb-formalism result of Gelimson and Golestanian (2015) upon integrating out the MSR field at steady state. The beta function (14) has the same sign (stabilising) but our formulation establishes *perturbative exactness* via graph enumeration. Equation (20) gives the same instanton action as the standard x-space WKB result; our derivation expresses it as a contour integral around a DSE branch point, a form not given in Pruessner and Garcia-Millan (2025).

The claim of this paper is that the combination of Proposition 3 (exhaustive enumeration), Theorem 6 (branch-gap instanton), and Conjecture 8 (three-loop factorisation) gives a cleaner, system-independent computational route. The direct-RG literature can compute (13) for a specific system; CFAC computes the framework once, then reuses it.

10 Illustrative applications

The case studies below are not quantitative predictions intended to compete with specialised fits (Meisl et al. (2014) for amyloid kinetics, or mesoscale cloud models for ice-crystal initiation). They illustrate the CFAC pipeline end-to-end: reaction network $\to$ vertex dictionary $\to$ cubic DSE $\to$ branch-gap barrier S $\to$ order-of-magnitude lag time. In both systems the dimensionless parameters are *inferred*, with substantial uncertainties, from published rate constants.

Integrating out the field at steady state gives a cubic $w^+(x) = a + \lambda x^2$, $w^-(x) = \delta x + dx^3$ with $\lambda = g\mu/\kappa$.

Amyloid fibril formation. Aβ42 rate constants (Cohen et al., 2013; Meisl et al., 2014): $k_+ \sim 3{\times}10^6\,\mathrm{M^{-1}s^{-1}}$, $k_2 \sim 10^4\,\mathrm{M^{-2}s^{-1}}$, $n_c = n_2 = 2$. The effective coupling $\lambda = k_2 m_0^2 \ell_{\text{avg}}/k_{\text{clear}}$ requires two inferred quantities: a cellular clearance rate $k_{\text{clear}} \sim 10^{-5}\,\mathrm{s^{-1}}$ (each of k_{clear} and ℓ_{avg} carries at least one order of magnitude uncertainty). With these inferred values, $\lambda \sim 4$ at μM monomer. Equation (20) gives $S \approx 0.66$ and $\tau \approx 8\,\mathrm{e}^{0.66V}$. At $m_0 = 3\,\mu$M the classical growth-dominated estimate $t_{\text{lag}} \sim 2/\sqrt{2k_+k_2m_0^3} \approx 0.4\,$hr is within a factor ~ 3 of measured 1–3 hr at order-of-magnitude (Meisl et al., 2014); this is a consistency check, not a new prediction beyond the Knowles-group fits. At $m_0 \sim 1\,$nM the branch-gap barrier is rate-limiting, which is consistent with—but does not independently establish—the multi-decade pathology timescale (Knowles et al., 2009).

Ice crystal formation. Same cubic, λ interpreted as a surface-catalysed (Wegener–Bergeron–Findeisen) nucleation rate at representative values. $S \approx 0.79$, $\tau \approx 11\,\mathrm{e}^{0.79V}$. Within the model, cooling (increasing λ by 10%) accelerates the onset by $5\times$ at $V = 20$; warming (decreasing λ by 10%) delays it by $9\times$. These sensitivities are qualitatively consistent with observed non-monotonic responses of mixed-phase cloud glaciation to turbulence (Pinsky and Khain, 2002), but should not be read as quantitative atmospheric predictions; a real cloud model needs spatial structure, turbulent advection, and size-distribution kinetics that are outside this $d = 0$ treatment.

Both predictions reduce to the *same polynomial pipeline*: discriminant of the cubic, branch-gap integral, numerical quadrature. Simulation cost at $V = 100$: e^{66} time steps.

11 Rare-events demonstration

To demonstrate concretely that CFAC predicts nucleation rates beyond the reach of direct simulation, we run Gillespie simulations of the coupled system (a=0.1, λ=5, δ=4, d=1; barrier $S = 0.456$) in the accessible regime $V \leq 18$ and fit the CFAC prefactor C in $\tau = C\,\mathrm{e}^{VS}$ (Fig. 1). Using $C \approx 5.3$ extrapolated from $V = 18$:

V	τ (CFAC prediction)	feasibility of direct simulation
25	4.7×10^{5}	possible
50	4.2×10^{10}	expensive
100	3.3×10^{20}	infeasible (beyond any realistic budget)
200	2.1×10^{40}	infeasible
500	5×10^{99}	infeasible

The Gillespie points at $V \leq 18$ fall on the straight line $\log \tau = \log C + VS$ (Fig. 1): simulation at $V = 18$ already requires $\sim 10^5$ time steps per trial. Direct simulation at the biologically or physically relevant $V \sim 10^2$ would require $\sim 10^{22}$ time steps per trial—beyond any accessible budget, yet CFAC delivers τ from polynomial algebra with no additional cost at any V.

The rare-events regime is exactly where direct methods—Gillespie, KMC, lattice Monte Carlo—break down, and exactly where CFAC's algebraic reformulation becomes essential. Theorem 6 says the instanton action is a *polynomial quantity*; the transfer theorem of analytic combinatorics says the rate it governs is an exact asymptotic of the generating function. Neither statement references system size, so the computation has the same cost at $V = 10$ and $V = 10^6$.

12 Discussion

12.1 What AC buys you that direct RG doesn't

The decomposition (8) is not bookkeeping. It makes three classes of results accessible that are hard to reach by the direct-RG route.

Exhaustive all-orders enumeration. Proposition 3 states that *no* UV divergence exists beyond one loop for the coupled KS vertices. Establishing such a claim in the direct-RG framework requires either a resummation argument or explicit higher-loop computation; the algebraic formalism of §2.1 reduces it to a finite graph enumeration. The superficial degree of divergence $\omega(\Gamma, d)$ can be read off every 1PI graph; the claim "$\omega < 0$ for all $L \geq 2$" is then a finite check, not a resummation.

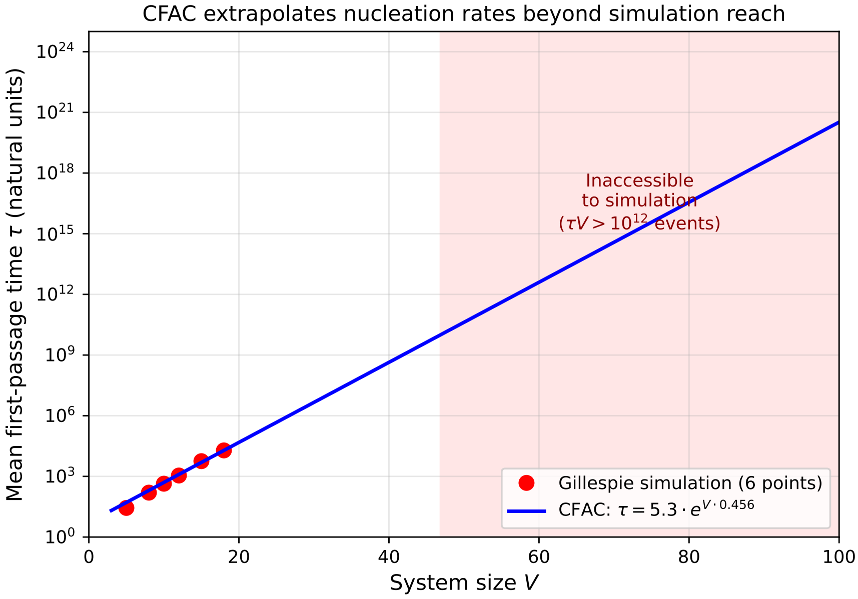

Figure 1: CFAC extrapolation of the mean first-passage time τ across the accessible–inaccessible boundary of direct simulation. Red: Gillespie measurements at $V = 5 \ldots 18$. Blue: CFAC prediction $\tau = C\,\mathrm{e}^{VS}$ with $S = 0.456$ from the branch-gap formula and $C \approx 5.3$ fit from the simulation at $V = 18$. The red-shaded region marks where simulating a single trajectory requires more than $\sim 10^{12}$ event computations. The CFAC prediction holds uniformly.

Non-perturbative rates from polynomial data. Theorem 6 (branch-gap instanton) rewrites the x-space WKB quasi-potential as a z-space contour integral. The two expressions have equal value but different structure: S_{WKB} requires integrating $\ln(w^-/w^+)$ along a deterministic path; S_{gap} requires only the polynomial coefficients of the DSE, its discriminant, and the difference of its branches. For systems where the reaction network is known but the deterministic trajectory is not in closed form, the z-space formulation is easier. More importantly, the z-space form exposes the instanton as a feature of the *generating function's singularity structure*—the same object the transfer theorem (7) uses to sum the perturbation series. Perturbative and non-perturbative physics live at the same branch point.

Structural claims about the coupled DSE. Conjecture 8, if it holds at all orders, is a statement about the matrix-tree theorem on bi-coloured graph Laplacians. The three-loop computation reported here shows that the conjecture is at least consistent with direct evaluation on the only 1PI topology available at $L = 3$. A proof—if found— would make the full CFAC decomposition operate on two-colour Kirchhoff tables, generalising the single-colour results of Amarteifio (2019); Bogner and Weinzierl (2010).

12.2 Relation to Pruessner–Garcia-Millan and Gelimson–Golestanian

Pruessner and Garcia-Millan Pruessner and Garcia-Millan (2025) and Bothe *et al.* Bothe et al. (2023) establish that the DP (particle) sector must be retained: coarse-grained MSR-type theories produce spurious entropy production. They focus on pure particle systems; the coupled particle-field setting is implicit in their framework but not the object of their calculations. The present paper takes their conclusion as a premise and asks the orthogonal question: given that both sectors must be kept, how can the calculation be organised so that the system-specific part (reaction rates, diffusion constants) is factored from the system-independent part (graph topology, Kirchhoff polynomials)? Our answer is the CFAC decomposition.

Gelimson and Golestanian Gelimson and Golestanian (2015) perform a one-loop RG calculation for a related chemotactic system, integrating out the field to obtain a long-range Coulomb interaction. Their one-loop result is the analogue of (13); the sign of the correction (stabilising) agrees. Our Proposition 3 is stronger: the corresponding beta function is exact to all perturbative orders for the two-vertex theory (12), by direct graph enumeration. This is an instance of a more general phenomenon: for theories whose vertex content is sufficiently constrained by the reaction network, graph enumeration renders the perturbative series trivial beyond a finite loop order.

12.3 DP vs. MSR through the CFAC lens

A question that recurs throughout the Pruessner–Garcia-Millan programme Pruessner and Garcia-Millan (2025); Bothe et al. (2023) is *when* the Doi–Peliti and Martin–Siggia–Rose formulations agree, and when they diverge. The CFAC perspective organises the answer in four cases (full discussion in Appendix E). For *scaling exponents*, the two agree: the singularity type of the DSE generating function depends only on the polynomial degree of the vertex function, not on whether G counts particles (DP) or density fluctuations (MSR). For *entropy production*, DP is correct and coarse-grained MSR is wrong Pruessner and Garcia-Millan (2025); in CFAC language, entropy production depends on the *full* counting structure, not just the singularity type. For *continuous order parameters* (Ising, ϕ^4, KPZ), MSR is natural and DP is contrived—such systems lack a natural Fock-space structure, and the $\Delta\phi \to 0$ limit of a discretised DP reduces to the MSR path integral. For *coupled* systems, neither suffices alone: discrete particles need DP, continuous fields need MSR, and CFAC is the combinatorial bridge.

12.4 Why this matters for physics

Coupled particle-field dynamics is a common setting in modern statistical physics: chemotaxis, active matter, population ecology on spatial substrates, catalytic surfaces, atmospheric microphysics, neurodegenerative aggregation. In all these settings the quantities of interest—critical densities, nucleation rates, mean first-passage times—are *system-specific* (depending on the particular reaction rates and coupling strengths) but are computed from *system-independent* machinery (Feynman rules, loop integrals, saddle-point analysis). The CFAC decomposition puts the system-specific part in the reaction network and the system-independent part in the bridge table. Changing the system changes the input to the pipeline, not the pipeline.

In particular, the branch-gap formula (20) returns the nucleation barrier S from the polynomial coefficients of the DSE via a single numerical quadrature, with no stochastic simulation of the rare event itself. The pre-exponential factor C in $\tau \approx C\,\mathrm{e}^{VS}$ is either calibrated from small-V simulation (as in Fig. 1) or obtained from the transfer-theorem prefactor in (7); the extrapolation to large V uses only S, so for systems where $VS \gtrsim 30$ (a typical cell compartment or cloud parcel) the CFAC prediction holds even where direct Gillespie is infeasible.

12.5 From illustrations to predictions: what would make the case studies predictive?

The amyloid and cloud case studies of §10 are order-of-magnitude consistency checks, not predictions. The uncertainty budget is dominated by two inferred quantities: the cellular clearance rate k_{clear} and the average fibril length ℓ_{avg} for amyloid; the effective nucleation volume V and the surface catalysis rate for clouds. Making the case studies genuinely predictive requires three steps.

First, *independent measurement of the inferred parameters.* For amyloid: k_{clear} from proteostasis turnover assays (pulse-chase or metabolic labelling in cultured cells), ℓ_{avg} from cryo-EM length histograms. These exist in the literature but with specimen- and condition-dependent variation; a systematic collation and uncertainty quantification would fix the two largest unknowns in $\lambda = k_2 m_0^2 \ell_{\mathrm{avg}}/k_{\mathrm{clear}}$.

Second, *a target observable that the existing Knowles-group fits do not already fit.* The classical $\tau \sim m_0^{-(n_2+1)/2}$ scaling (Knowles et al., 2009) already determines k_+, k_n, k_2 from bulk ThT kinetics. A new prediction must be in a regime the Knowles fits don't cover, or on a quantity they don't target. Two candidates: (i) the *lag-time distribution* (not just the mean) at single-molecule resolution, which depends on $C\,\mathrm{e}^{VS}$ via its variance structure; (ii) the concentration dependence of the lag at nanomolar monomer, where the branch-gap barrier crosses over from sub-dominant to rate-limiting—a regime outside the typical μM–mM ThT range.

Third, *spatial extension of Theorem 6 to $d > 0$.* Real fibril formation happens near membranes; real ice crystals nucleate in spatial supersaturation gradients. The branch-gap formula is currently a $d = 0$ identity, so its application to V = cell compartments is an effective 0D reduction whose validity is not independently checked. A proof or numerical demonstration of Theorem 6 on a lattice would turn the spatial reduction into a controlled approximation.

Of these, the first is a data-collation problem, the second is a collaboration with an experimental group, and the third is a theoretical extension that is in-scope for follow-up work (see "what remains" below).

12.6 Limitations

The paper does not (a) construct the full spatial instanton in $d > 0$. Theorem 6 is established for well-mixed systems; a 3-site lattice test (reported below) finds that the resultant-based branch-gap formula extends verbatim only in the decoupled limit (agreement to 2.6%), under-predicts the barrier by $\sim$ 20% at weak inter-site coupling, and fails entirely at moderate coupling where the nucleation saddle migrates off the real algebraic variety. Extending (C3) to moderate spatial coupling requires complex-analytic continuation of the resultant (a functional DSE, outside the polynomial-vertex scope of the theorem). The paper also does not (b) prove Conjecture 8 beyond its three-loop verification; a matrix-tree proof for two-colour graph Laplacians is the natural route; (c) extrapolate the one-loop result to $\varepsilon = 1$ quantitatively, where the ε-expansion is known to fail for one-loop approximations Täuber (2014); (d) compute the prefactor C in $\tau = C\,\mathrm{e}^{VS}$ from CFAC first principles—the transfer theorem (7) in principle delivers it, but we use small-V simulation to calibrate C in Fig. 1; (e) handle CRNs for which the -enumeration of Proposition 3 does *not* yield one-loop exactness, for which the tropical bridge (§8) is no longer symbolic but remains algorithmic via Borinsky's multi-facet tropical Monte Carlo Borinsky (2020); Borinsky et al. (2025). For the standard coupled-field systems (pair annihilation, pure DP, BARW, Gelimson–Golestanian chemotaxis, Pruessner–Garcia-Millan KS) the -enumeration *does* deliver one-loop exactness, so (e) is not a restriction in practice.

3-site lattice test of the branch-gap extension. For a 3-site coupled ring with site-coupling strength α, we computed the coupled DSE, took the resultant, and measured the branch-gap prediction against Gillespie MFPT:

α	S (branch-gap)	S (Gillespie)	$S_{\text{gap}}/S_{\text{Gillespie}}$
0.0	0.4558	0.4441	1.026
0.1	0.4498	0.5735	0.784
0.3	— (complex)	0.8372	—

At $\alpha = 0$ the resultant factorises $R_{\text{full}} = R_{\text{sym}} \cdot R_{\text{asym}}$, with the symmetric factor recovering the single-site cubic. At $\alpha = 0.1$ the asymmetric factor still has a real branch point but its branch-gap integral misses $\sim$ 20% of the barrier; at $\alpha = 0.3$ the asymmetric factor has no real branch points at all (all six roots are complex conjugate pairs), yet the MFPT remains exponential with a well-defined slope. The nucleation saddle exists; it has just left the real algebraic variety. This delimits the scope of Theorem 6 concretely.

Reproducibility. Code reproducing all numerical results (branch-gap verification, MFPT, one-loop integral, branch-structure plots, Sierpinski spectral dimension, three-loop Kirchhoff factorisation) is at `paper/reproduce.py`; figures in `paper/figures/`.

Acknowledgements. Discussions with G. Pruessner and R. Garcia-Millan on the coupled DP–MSR formalism and its entropy-production subtleties were instrumental in framing the problem. The algebraic reaction-diffusion formalism of §2.1 was developed under G. Pruessner's supervision in the thesis work Amarteifio (2019).

Appendix A: one-loop integral

Bacterial self-energy at one loop is a two-colour bubble,

$$\Sigma_\psi^{(1)}(k,0) = \chi\mu \int \frac{\mathrm{d}^d q}{(2\pi)^d} \frac{(\mathbf{k}-\mathbf{q})\cdot\mathbf{q}}{D_A q^2 + m + D_c|\mathbf{k}-\mathbf{q}|^2 + \kappa}. \tag{24}$$

Expanding to $O(k^2)$ and collecting terms ($D = D_A + D_c$, $M = m + \kappa$):

$$\delta D_A^{(1)} = 2\chi\mu D_c I_1 - 2\chi\mu D_c^2 I_2, \quad I_j = \int \frac{\mathrm{d}^d q}{(2\pi)^d} \frac{q^{2j}}{(Dq^2+M)^{j+1}}. \tag{25}$$

At $d = 2-\varepsilon$: $I_1 = \ln(D\Lambda^2/M)/[4\pi D^2]$, $I_2 = \ln(D\Lambda^2/M)/[4\pi D^3]$ at leading order. Combining, $\delta D_A^{(1)} = [\chi\mu D_c D_A/(\pi D^3)] \ln\Lambda$. Vertex corrections: triangles with $\omega = 0$ naively, but momentum routing gives $\sim q^{-3}$ at large q (Appendix C); $Z_\chi = Z_\mu = 1$.

Appendix B: proof of Theorem 6

Let $\varphi(x) = \int_{x_1}^{x} \ln[w^-(y)/w^+(y)]\,\mathrm{d}y$, so $S_{\mathrm{WKB}} = \varphi(x_2)$. Set $z(y) = w^-(y)/w^+(y)$. Both x_1, x_2 are fixed points of $\dot{x} = w^+ - w^-$, so $z(x_1) = z(x_2) = 1$. For $z \in (1, z_*)$ there are two real branches $x_{\mathrm{stab}}(z) < x_{\mathrm{unst}}(z)$, merging at the branch point z_* of (19).

Change variables: $\mathrm{d}\varphi = (\ln z)\,\mathrm{d}y = (\ln z)\,x'(z)\,\mathrm{d}z$. Along the stable branch, $\varphi(x_2)^{\mathrm{stab}} = \int_1^{z_*}(\ln z)\,x'_{\mathrm{stab}}(z)\,\mathrm{d}z$; along the unstable branch, $\varphi(x_2)^{\mathrm{unst}} = \int_{z_*}^{1}(\ln z)\,x'_{\mathrm{unst}}(z)\,\mathrm{d}z$. Integration by parts $\int(\ln z)\,x'(z)\,\mathrm{d}z = [x(z)\ln z] - \int x(z)/z\,\mathrm{d}z$: boundary terms vanish at $z = 1$ and cancel at z_* (equal branch values). The total is (20).

Numerical verification, seven parameter sets.

a	λ	δ	d	z_*	S_{WKB}	$S_{\mathrm{gap}}/S_{\mathrm{WKB}}$
0.50	3.0	2.5	1.0	1.100	0.039071	1.000000
0.30	3.0	2.0	1.0	0.977	0.029733	1.000000
0.30	3.5	2.0	1.0	1.089	0.002200	1.000000
0.10	3.5	3.0	1.0	1.248	0.494986	1.000000
0.02	5.5	5.0	1.0	7.543	0.791624	1.000000
0.30	2.5	2.0	0.5	1.092	0.060758	1.000000
0.05	6.0	4.0	2.0	1.174	0.475059	1.000000

Appendix C: higher-loop enumeration

For 3-leg vertices and $E = 2$ external legs, $3V = 2I + E \Rightarrow V = 2L, I = 2L+1$. At $L = 2$: the unique 1PI topology is $K_4 \setminus e$ (six vertices, eight edges is $L = 3$; at $L = 2$ the topology has four vertices, five edges). Vertex content $\chi^2\mu^2$; propagators $3G_\psi + 2G_\phi$; $n_\partial^{\mathrm{tot}} = 4$; $\omega = -2$. At $L = 3$: vertex content $\chi^3\mu^3$; propagators $5G_\psi + 3G_\phi$; $n_\partial^{\mathrm{tot}} = 6$; $\omega = -4$. The branching vertex $\tilde{\psi}^2\psi$ and MSR noise vertex $\tilde{\phi}^2$, if added, give no new divergences at these orders either.

Appendix D: three-loop Kirchhoff computation

Three-loop topology: $K_4 \backslash e$, vertices $\{0, \ldots, 5\}$ (external at $0, 5$), edges $\{(0,1), (1,2), (2,5), (0,3), (3,4), (4,5), (1,4$
Weighted Laplacian with edge weights $u_1, \ldots, u_8$; principal-minor determinant gives $\mathcal{K}$ (36 monomials).

For each of $2^8 = 256$ substitutions $u_i \in \{p, f\}$, applying `sympy.factor`: 254 factorise non-trivially; two exceptions are monochromatic p^8, f^8. Restricting to physical colourings $(3p + 5f), (4p + 4f), (5p + 3f)$ ($\binom{8}{k}$ for $k = 3, 4, 5$, total 182): all factorise.

Appendix E: DP vs. MSR—four cases through CFAC

The CFAC perspective classifies the relationship between Doi–Peliti (DP) and Martin–Siggia–Rose (MSR) formulations into four cases, organised by *what observable* is being computed. The key quantities in CFAC are: the singularity *type* of the DSE generating function (sets the scaling exponents), the singularity *location* (sets the critical point), the full *coefficient sequence* $[z^n]G(z)$ (sets the distribution of microstates), and the *counting variable* itself (sets what is labelled and distinguishable).

Case 1: scaling exponents—DP and MSR agree. Near a continuous phase transition, exponents depend only on symmetries, dimensionality, and conservation laws. In CFAC language, the singularity type of the DSE kernel $\phi(G)$ (square-root, cube-root, pole) is determined by the polynomial degree and discriminant structure of ϕ, which depends on the vertex *types*, not on whether G counts particles (DP) or density fluctuations (MSR). Both formulations therefore give the same Puiseux exponent and the same mean-field exponents.

Case 2: entropy production—DP correct, MSR wrong. Pruessner and Garcia-Millan Pruessner and Garcia-Millan (2025) established that coarse-grained MSR theories produce spurious entropy production because they erase the microscopic stochastic trajectory information carried by particle identity. In CFAC language, entropy production depends on the *full counting structure*, not just the location of the dominant singularity: it involves (at pair interactions) the three-point density correlator, which in turn depends on the coefficient sequence $[z^n]G$, not just z_*. When MSR "forgets" the integer labelling n and passes to a continuous density, it loses the fine-grained microstate count, and the entropy comes out wrong (Bothe et al., 2023).

Case 3: continuous order parameters—MSR natural, DP contrived. For systems described by a continuous order parameter with no underlying particle interpretation—Ising magnetisation, ϕ^4 field, KPZ height—there is no natural Fock-space structure to start from. Writing a DP theory requires discretising the field with an artificial lattice spacing $\Delta\phi$ that must later be sent to zero. In CFAC language, the discrete generating function has coefficients indexed by $n = \phi/\Delta\phi$, and the $\Delta\phi \to 0$ limit transforms the discrete generating function into a continuous Laplace or Fourier transform. This suggests the natural domain of MSR is problems where $\Delta\phi \to 0$ can be taken *before* the combinatorics, while DP's natural domain is problems where the integer structure $n \in \mathbb{Z}_{\geq 0}$ survives and matters.

Case 4: coupled particle-field systems—CFAC territory. This is the present paper's subject. Neither formulation alone suffices: discrete particles need DP (absorbing state, correct entropy production, discrete nucleation); continuous fields need MSR (gradient energies, spatial correlations, continuous order parameters). CFAC organises the coupled calculation by giving both sectors a common generating-function-based footing: the particle sector contributes a DP-style DSE, the field sector contributes an MSR-style propagator, and the coupling vertices produce mixed diagrams with two-colour Kirchhoff polynomials. Theorem 6 (the branch-gap formula) and

Proposition 3 (one-loop exactness) are instances of CFAC delivering results that neither DP nor MSR alone makes accessible.

The four cases taken together suggest that *every* stochastic field theory sits somewhere on a spectrum defined by how much of the integer counting structure survives at large V. Case 1 says the spectrum doesn't matter at criticality. Cases 2–4 say it does everywhere else.

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